# UI Guide — Duolingo Style Quest Study Application

A standalone guide for building the React frontend in the one UI style the team picked — intended for an agent in an integrated development environment (IDE). Read it together with the Technical Specifications, which remain the authority for the stack, the API and the database.

## 1. How to use this guide

The IDE build so far is backend only: Django and Django REST Framework, PostgreSQL 16 or later, and local Ollama (Qwen2.5 3B Instruct). This guide covers the next build — the React frontend — and only the frontend. The backend stack, the planned REST API and the database design do not change; they are specified in the Technical Specifications document.

The style specified here is the Duolingo style from the Quest Study UI Styles comparison page, the direction the team picked. No other compared style is part of this build. Do not mix in elements from the other styles.

- **The mock is a reference, not code:** the comparison page is a static mock with sample data. Do not copy its HTML into the app. Build real React components that reproduce its look, layout and motion, with all data coming from the planned REST API.
- **Mock files are delivered with this guide:** the duolingo-mock folder contains a standalone Duolingo-only page (index.html) with five screens — Student Dashboard, Quest Board, Quest Detail, Teacher View and Boss Fight — plus the design tokens (tokens.css and tokens.json), the boss sprite (boss-golem.svg) and the icon set (icons.svg). Read the exact values from those files instead of estimating them from a screenshot or a PDF.
- **Where the mock and this guide differ, this guide wins:** for example, the profile streak calendar in Section 5.5 is specified here and does not appear in the mock.
- **Sample data is sample data:** values such as Level 7, 1,240 / 1,400 XP, a 12-day streak and the Limit & Turunan boss quiz exist only to make the mock comparable. Never hardcode them in the app.

## 2. Locked scope for the frontend

Build exactly this scope. The exclusions matter as much as the inclusions: do not add screens, buttons or placeholders for anything outside it.

- **Audience:** senior high school students, plus a teacher view. The tone is playful but not childish — game mechanics and artwork, no mascot costume.
- **Two quest types:** Assignment Quests (set by the teacher, deadline-driven) and Revision Quests (set by the student, exam-preparation-driven), shown with the same card component and separated by filter tabs.
- **Teacher-lite:** teachers create quests and boss quizzes and view students' progress and quiz scores. There is no file submission by students and no file grading by teachers — include no upload, attachment or grading controls anywhere in the UI.
- **Full CRUD through the API:** quests and quizzes are created, read, updated and deleted through the REST API. The browser never accesses PostgreSQL directly and never calls Ollama directly.
- **Gamification:** XP, a level-up every 200 XP, streaks and badges. Nothing else: no multiplayer, no leaderboard and no shop — do not add UI for them, not even as disabled placeholders.

- **Regular quest quiz:** a quest may have a quiz as one of its steps (Section 5.3). It has 5 questions, no timer, and a pass line of 80% (4 out of 5). Not passing leaves that quiz step unchecked and the quest incomplete. Retry is allowed and the best attempt counts.
- **Boss exam:** the boss fight is an exam, not practice (Section 5.4): 20 questions, a 1-hour timer, boss and player HP, a real fail state, and at most 2 retries after losing (3 attempts total).
- **Quiz generation and caching:** the teacher generates a quiz once per session through the server-side AI endpoint; the result is cached server-side and the same cached quiz may be reused for a different class on the same subject. The frontend never generates questions itself.
- **Responsive web app:** one React app that works well in both phone and desktop browsers. No separate mobile app and no phone-only or desktop-only screens.

## 3. Design tokens

Define every value in this section once — as CSS variables or a single theme object — and consume the tokens from components. Components must not hardcode hex colours, radii or shadows. Values are copied from the mock's Duolingo theme; tokens.css and tokens.json in the duolingo-mock folder carry the same values in machine-readable form.

### 3.1 Colours

| Token | Value | Used for |
|---|---|---|
| --mb (background) | #ffffff | App background |
| --ms (surface) | #ffffff | Cards and panels |
| --ms2 (tint surface) | #f2f9e9 | Soft green tint: selected nav item, answer tiles, path trail |
| --mt (text) | #3c3c3c | Main text |
| --mm (muted text) | #777777 | Secondary text, labels, due dates |
| --ma (primary) | #58cc02 | Primary buttons, XP bar fill, active tab, completed states |
| --ma-dark | #46a302 | Bottom edge of primary buttons (hard shadow) |
| --ma-ink | #ffffff | Text and icons on the primary colour |
| --ma2 (secondary accent) | #ff9600 | Streak flame and streak number, current-question dot, avatar disc |
| --mbord (border) | #e0e0e0 | Card borders and hard shadows |
| Boss card | #ce82ff, border #a560e8 | Boss teaser card on the dashboard; its inner bar track is #a560e8 and its sub-text is #f3e4ff |
| Danger / boss face | #ff4b4b, edge #d33131 | Boss icon disc, Start Boss Fight button family, error feedback |
| HP fill | #e5484d | Boss HP bar fill |
| Gem filled / empty | #38bdf8 / #94a3b8 | Difficulty gems; the empty gem also renders at 22% opacity |
| Status: Not started | #f4f4f4 bg, #777777 text, #e0e0e0 border | Quest status pill — not started yet |
| Status: In progress | #fdf0dd bg, #92580a text, #f3ddba border | Quest status pill — started, not finished (same family as the XP tag) |
| Status: Done | #e7f6ec bg, #157347 text, #c2e8cf border | Quest status pill — completed (same family as the Revision tag) |

## 3.2 Subject colours

Each subject always keeps its own colour and icon (Section 6). A quest's colour comes from its subject, never from its type or status.

| Subject | Colour | Icon |
|---|---|---|
| Matematika | #2563eb (blue) | Sigma (Σ) |
| Biologi | #16a34a (green) | Leaf |
| Fisika | #d97706 (amber) | Atom |
| Bahasa Inggris | #db2777 (pink) | Open book |

## 3.3 Typography

| Element | Specification |
|---|---|
| Heading font | Fredoka, weights 500–700, sans-serif fallback. Screen titles, section headings, the logo and the boss name. This rounded typeface is a core part of the style — do not substitute a generic sans-serif for headings. |
| Body font | Nunito, weights 600–900, sans-serif fallback. All body text, cards, buttons and labels. |
| Base size | 14.5px body text in the app shell; quest titles 0.98rem at weight 800; question text 1.14rem at weight 700 with 1.45 line height. |
| Stat values | 1.22rem at weight 900 in this style (the mock raises the generic 800 to 900 for the Duolingo theme). |
| Labels | 0.68–0.74rem, weight 800, uppercase, letter-spacing 0.07em, muted colour — stat labels, question kicker, table headers. |
| Buttons | Uppercase text with letter-spacing 0.03em (primary, create and check buttons in this style). |
| Numbers | Tabular numerals for the boss timer, HP values and XP figures so digits do not shift width while counting. |

Load Fredoka and Nunito from Google Fonts, as the mock does, with the sans-serif fallbacks above so the app still reads correctly if the fonts fail to load.

## 3.4 Shape, spacing and shadows

| Property | Value in this style |
|---|---|
| Corner radius | 22px on cards, panels and the app frame (the mock's Duolingo theme raises the base 18px to 22px). Path nodes 20px; subject tiles 13px; buttons 10–11px; pills and bars fully rounded (999px). |
| Borders | 2px solid #e0e0e0 on cards, stat tiles, question and answer cards. This style uses visibly chunky borders, not hairlines. |
| Shadows | No blurred drop shadows on cards. Cards sit on a hard offset shadow: box-shadow 0 3px 0 #e0e0e0. Path nodes and subject dots use 0 4px 0 in the border colour or rgba(0,0,0,.2). Speech bubbles use 0 3px 0 rgba(0,0,0,.12). |
| Progress bars | 15px high in this style (base mock: 10px), fully rounded, track in the tint or border colour, fill in the primary colour; the boss HP bar is 13px high with the HP red fill. |
| Spacing | Screen body padding 18px; vertical rhythm 16px between sections; 14px between path nodes; 9–10px inside card rows and grids; card padding 12–18px depending on the card. Keep touch targets at least 44px high. |

### 3.5 Buttons

| Button | Style and behaviour |
|---|---|
| Primary (Start, Check Answer, Create Quest) | Primary green background, white uppercase text, radius 10–11px, hard bottom edge box-shadow 0 4px 0 #46a302. On press, the button moves down 2px and the shadow shrinks to 0 2px 0 — the mock's chunky press feel. |
| Ghost (View, Create Boss Quiz) | White or tint background, main-text colour, 1.5–2px border in the border colour, no hard shadow. |
| Boss action (Start Boss Fight) | Red family: #e5484d to #ff4b4b background, white text, weight 900. Reserved for entering and progressing the boss fight so it always stands apart from study actions. |
| Disabled | Muted tint background and muted text, no shadow, no press movement. Use for unavailable actions only — never for out-of-scope features (Section 2). |

## 4. Shared layout and components

Every screen is built from the same shell and components, so the game layer looks identical wherever the student is.

- **App shell:** white background; a top bar with the Quest Study logo (shield crest in a green rounded square), navigation (Dashboard, Quests, Profile) with the active item in green on the tint surface, and the student's initial in an orange avatar disc. On phones the same three destinations move to a bottom navigation bar with icons and labels; the logo bar stays on top.
- **Content column:** a single centred column, maximum about 880px wide on desktop, full width with 18px side padding on phones. The teacher progress table may widen to about 1,100px on desktop (Section 5.6).
- **Stat strip:** four stat cards in a row — Level, XP, Streak, Daily goal — each a 2px-bordered card with an uppercase label and icon, a large weight-900 value and a small muted sub-line. The XP card and the Daily goal card carry the 15px progress bar. The streak value is orange with the flame icon. Data comes from GET /api/progress/me/ (Section 9).
- **Quest card anatomy:** subject tile or path dot (subject colour and icon), title, then a meta row: difficulty gems with the difficulty word, an XP pill, a type tag (Assignment: #e8effc background, #1d4fa1 text, #c9dbfa border; Revision: #e7f6ec background, #157347 text, #c2e8cf border), the due date with a clock icon and a status pill (Section 5.2). The whole card is the tap target to the quest detail (Section 5.3); on the dashboard path the action button shows Start, or Done in the completed state.
- **Calm screens, dramatic boss:** the quest board and quest detail (Sections 5.2–5.3) deliberately reuse these tokens without the boss fight's drama — no timer on study screens or on the regular quest quiz, no HP bar, no damage numbers, no arena. The 1-hour timer, both HP bars and the fail state exist only in the boss exam (Section 5.4). Keep study screens calm and scannable so the boss exam still feels like an event when it arrives.
- **Boss teaser card:** the purple boss card (Section 3.1) that appears on the dashboard: boss stage strip with the golem sprite and a speech bubble, boss name and topic, the exam facts (20 questions, 1 hour, boss 100 HP), the boss HP bar, XP reward and a Start Boss Fight button.

## 5. Screens

### 5.1 Student dashboard — the winding learning path

The dashboard is a path, not a list. From top to bottom:

- **Hero row:** the student hero sprite (64px, gentle float animation, Section 7) beside the greeting and one short line about today's goal. Keep the hero small — it is a companion, not a mascot takeover.

- **Stat strip:** the shared four-card strip from Section 4.
- **Today's Path:** today's quests as nodes on a winding trail. A vertical dashed trail (8px wide, repeating tint dashes) runs behind the nodes. Each node is a card 88% of the column wide, offset alternately 14px left and 14px right of centre, with a 52px circular subject dot (subject colour, white subject icon at 26px, hard shadow 0 4px 0 rgba(0,0,0,.2)) overlapping its left edge. Node content matches the quest card anatomy in Section 4, with the action button showing Start, or Done once the quest is complete.
- **Boss teaser:** the purple boss card from Section 4 for the next available boss quiz.
- **Badge row:** the student's most recent badges as medallions (Section 6), linking to the profile.

Order the path by due date, nearest first, with completed quests staying visible in place for the day. If there are no quests today, show an empty state in the same style — hero sprite, one friendly line, and a button to the quest board — never a blank area.

### 5.2 Quest board — calm and scannable

The board is where all quests live; the dashboard path is only today's slice of it. It is a regular study screen, not a battle: the same Duolingo tokens, cards and icons as everywhere else, arranged for scanning. There is no timer, no HP bar, no boss sprite and no damage feedback on this screen — the only pressure a quest carries here is its deadline. It reuses the quest card in a simple vertical stack (no winding offsets) so long lists stay readable.

- **Header:** the screen title in Fredoka, one short line with the counts that matter (for example: 4 quests · 1 due today · 2 done), and the shared stat strip is not repeated here — the board starts with its filters so the list is visible without scrolling on a phone.
- **Filters — three groups, all working:** type (All / Assignment / Revision), subject (All plus one chip per subject, with the subject icon) and status (All / Not started / In progress / Done). Filters combine: choosing Revision + Biologi + In progress shows only quests that match all three. Every filter maps directly to the API: GET /api/quests/ with type=assignment|revision, subject and status parameters, refetched when a filter changes — never filter a hardcoded list in the browser. The active chip is green with white text; inactive chips are white with a 2px border and muted text. A result count line updates with the filters (for example: Showing 2 of 4 quests).
- **Quest card — every card shows all of this:** the subject tile (46px, subject colour and icon, Section 6.1); the title; the type tag (Assignment or Revision); difficulty as 1–3 gems with the difficulty word next to them (Easy 1, Medium 2, Hard 3 — never gems or colour alone); the XP reward as an XP pill; the deadline with a clock icon, with an overdue or due-today deadline also stated in words, not colour alone; and a status pill: Not started (tint background, muted text), In progress (amber family: #fdf0dd background, #92580a text, #f3ddba border) or Done (green family: #e7f6ec background, #157347 text, #c2e8cf border). Status comes from the quest's status field in the API response.
- **Opening a quest:** the whole card is one tap target and opens the quest detail (Section 5.3). Do not put a separate Start battle, timer or fight action on the card — starting and finishing a quest happens in the detail. On the card itself the only secondary action is the student's own Revision Quest menu (edit, delete) where the API allows it.
- **Student actions:** create, edit and delete their own Revision Quests with a form (title, subject, difficulty, deadline), reusing QuestForm (Section 9.1). Assignment Quests are read-only for students: no edit or delete controls on those cards.
- **Sorting and states:** default order is due date, nearest first, with done quests sinking below unfinished ones only when the status filter is All. Provide loading, empty (per filter combination, with a Clear filters action) and error states in the same card styling; an error state offers a retry button and never shows raw API errors.

### 5.3 Quest detail — finish the study, unlock the boss

Tapping a quest card opens its detail. This is the regular quest screen: one quest, its study material and a clear finish line. It shares the boss fight's visual language — same tokens, same cards, same reward feedback — but it is

not a boss fight with the boss removed. There is deliberately no timer, no HP bar, no enemy and no damage anywhere on this screen. The ordinary study steps cannot be failed, only finished; the one exception is the quiz step (below), which must be passed for the quest to complete. From top to bottom:

- **Back and header:** a ghost Back button to the board (keeping the board's filters), then a header card: the subject tile at full size, the quest title in Fredoka, the subject name and the type tag, and the status pill from Section 5.2. The header shows the same status the board showed — the two screens never disagree.
- **Reward and deadline row:** three small stat-style tiles in the shared card styling: XP reward (bolt icon, for example 100 XP), deadline (clock icon, date plus days remaining or Due today / Overdue in words) and difficulty (gems plus the difficulty word). No other numbers compete with these three.
- **Description:** the quest's description from GET /api/quests/{id}/ as plain readable body text — what to study and what done looks like. If the API returns no description, show the teacher's or student's short summary line instead of an empty gap; never invent study content in the frontend.
- **Steps checklist — study steps:** the quest's ordinary steps as a small checklist, one row per step: a 24px chunky checkbox (2px border, radius 7px; green fill with a white check when ticked, exactly the dashboard check styling), the step label, and nothing else. Ticking and unticking a study step is free and reversible, updates the steps progress bar under the checklist and moves the quest to In progress on the first tick. Study-step state is saved through PATCH /api/quests/{id}/ with the quest's progress and status; on reload the checklist shows the saved state, not a fresh one. Study steps are self-reported — nobody grades them.
- **Steps checklist — the quiz step is different:** where a quest has a quiz, the quiz is one of its steps and it is not self-ticked. Its row shows a quiz icon, the label (for example: Quest quiz · 5 questions · pass 4 of 5), the student's best score so far if they have attempted it, and a Start Quiz / Retry Quiz button that opens the regular quest quiz (Section 5.3.1). The quiz step's checkbox fills only when the server confirms a passing attempt. A student must never be able to tick the quiz step by hand, and a failed attempt leaves it unchecked.
- **Complete Quest button:** one primary button, full width on phones, labelled Complete Quest with the XP it awards (for example: Complete Quest · +100 XP). For the ordinary study steps the checklist is a study aid, not a gate: the button stays available whether or not every study step is ticked. The quiz step is the exception — if the quest has a quiz step that has not been passed yet, the button is disabled with one line explaining why (Pass the quest quiz to complete this quest) and the quest stays incomplete. Once the quiz is passed, the button enables. There is no second competing action next to it.
- **Completing a quest — the reward pop:** on completion the screen answers with the same kind of reward moment as a boss hit, in study colours: a reward pop pill rises and fades over the header card (+100 XP · Quest complete, green on white, 1s, the damage-pop motion from Section 7 run in reverse meaning — a gain, not a loss), the XP bar in the header area animates its width to the new total over about 0.4s, the status pill flips to Done, the checklist locks with every step shown ticked, and the Complete Quest button is replaced by a completed state (Done · +100 XP earned). All values — the XP awarded, the new total, the new level if the student levelled up — are the values returned by the completion request (PATCH /api/quests/{id}/ to the done status) and a refreshed GET /api/progress/me/; the client animates them, it does not calculate the official total itself. If the completion request fails, nothing on the screen changes state and an error line with a retry action appears — never show a reward for a completion the API did not confirm.
- **Boss unlocked state:** when the completed quest is the study material for a boss quiz, completing it unlocks that boss — this link is the core loop of the app, so it gets its own moment. Below the completed state, show a Boss unlocked banner in the boss card's purple family (#ce82ff background, #a560e8 border): the small golem sprite (84px, idle animation), the boss name and topic, one line (You finished the study material — the boss quiz is open), and a red Fight the Boss button that goes straight to that boss fight (Section 5.4). Whether a boss is unlocked, and which quiz it is, comes from the API (the quest detail's linked quiz and GET /api/quizzes/); before completion the same place on the screen shows the locked version in muted styling (Boss locked · Finish this quest to unlock Limit & Turunan) with no button. A quest with no linked boss quiz shows no banner at all — never a disabled placeholder for a boss that does not exist.

- **Read-only cases:** a done quest reopens in its completed state (locked checklist, earned XP shown, boss banner unlocked if it has one). An Assignment Quest's detail is fully completable by the student — teacher-set does not mean teacher-completed — but its description and steps are not editable by the student.

### 5.3.1 Regular quest quiz — low-stakes practice that still checks answers

The regular quest quiz is a normal quiz, presented in the same question-card styling as the boss exam but without any battle framing. It is how the app checks understanding inside a quest, before the boss exam.

- **Shape:** exactly 5 questions, one on screen at a time, four answer options (A–D), a Check Answer action, instant right/wrong feedback, and a progress indicator (Question 2 of 5). No timer is shown or enforced — the timer exists only in the boss exam.
- **Pass line:** 80% — 4 out of 5 correct passes. The result screen states the score as a count and a percentage (for example: 4 / 5 · 80% · Passed) so the pass is never a colour alone.
- **Effect on the quest:** passing ticks the quiz step in the quest detail (Section 5.3) and allows the quest to be completed. Not passing leaves the quiz step unchecked and the quest incomplete, however many study steps are ticked.
- **Retry and best attempt:** retry is always allowed on the regular quiz, with no attempt limit. The score the quest, the teacher view and the progress data use is the best attempt, as returned by the API — a later worse attempt never lowers it. Show the best score on the quiz step row and on the result screen.
- **No battle language:** no HP, no damage, no streak bonus, no win/lose panel. Feedback is study feedback: correct / not quite, plus a short hint at the method. The reward for passing is the ticked step and quest completion, not combat XP.

## 5.4 Boss fight

A boss fight is an exam presented as a battle — the high-stakes counterpart to the low-stakes regular quest quiz (Section 5.3.1). One question is on screen at a time, and the fight can genuinely be lost. The full rules are in Section 5.4.1; the screen itself, from top to bottom, is:

- **Arena:** a stage panel (16px radius, 2px border) with a sky-to-ground scene in the mock — sky #8ed6ff down to 62% of the height, ground #7ac74f, and a soft sun glow #ffe9a8 at the bottom centre. The Granite Golem sprite stands centre stage at 168px with its name pill underneath and, in this style, a white speech bubble above it with a taunt line (14px radius, hard shadow). A damage pop element sits ready at the top centre (Section 7).
- **Fight header:** boss icon disc (red, crossed-swords icon), boss name and topic in Fredoka, the question counter (Question 3 of 20), the attempt counter (Attempt 1 of 3, with retries left), and the countdown timer in a pill with tabular numerals. The timer starts at 60:00 (1 hour). This is the only timed screen in the app.
- **HP bars — both fighters:** two labelled bars, always with numbers as well as fill: Boss HP 100 / 100 (red fill) and Your HP 100 / 100. Next to them, the live streak indicator (for example: Streak × 4) and the XP earned so far in this attempt. Both HP values, the streak and the XP come from the attempt state returned and confirmed through the API; the client animates changes, it does not decide the official outcome.
- **Question card:** a 2px-bordered card with an uppercase kicker (Question 3 · Limits) and the question at 1.14rem. Below it, four answer buttons in a two-column grid (one column on phones), each with a letter tile (A–D). The selected answer takes the tint background and a green border with a green letter tile.
- **Check flow:** the student selects an answer, then presses Check Answer; each question is answered once and the fight then moves on — there is no re-trying the same question inside an attempt. Correct: the golem plays its hit animation, the damage pop shows -10 DAMAGE, boss HP drops by 10, and the feedback line confirms the hit, the +5 XP, and any streak bonus. Wrong: the streak resets, the feedback line gives a short hint at the method and states the consequence plainly — first wrong answer in a row costs nothing; a further consecutive wrong answer costs 10 player HP, shown as the player's bar dropping. A progress indicator under the answers shows answered, current and upcoming questions across all 20.

- **Victory panel:** when boss HP reaches zero, the golem plays its defeat crumble and a victory panel shows the correct count, the XP breakdown (per-answer XP, streak bonus XP, and the +100 XP win bonus as separate lines), any badge earned, and a button back to the dashboard.
- **Defeat panel:** on any loss (Section 5.4.1) a defeat panel states why the attempt failed in words — your HP reached zero, the 1 hour ran out, or the 20 questions ran out with the boss still standing — plus the score so far and, while retries remain, a Retry Boss button with the attempts left (for example: Retry Boss · 2 retries left). When no retries remain, the panel says so and offers only the route back; there is no fourth attempt.
- **Submission:** every finished attempt — win, loss or time-out — is submitted with POST /api/quizzes/{id}/attempts/ and the result screens read the returned score, XP and outcome. The client never calculates the official score, HP outcome or XP itself.

### 5.4.1 Boss exam rules — the exact numbers

Build these numbers exactly as written. They are the exam; do not soften the fail state, add extra lives, or change a number in the frontend.

| Rule | Specification |
|---|---|
| Questions | 20 questions per attempt. |
| Randomisation | Question order is randomised and the answer options of each question are shuffled, both per attempt, decided server-side when the attempt starts. Two students in the same class, and two attempts by the same student, should not see the same order. Randomisation reduces copying; it does not promise cheating is impossible, and the UI must not claim otherwise. |
| Timer | 1 hour (60:00), counting down only during the boss exam. No timer exists on the regular quest quiz or on any study screen. When the timer reaches zero the attempt ends and counts as a fail, whatever the HP state. |
| Boss HP and damage | Boss has 100 HP. Each correct answer deals 10 damage, so 10 correct answers defeat the boss. |
| Player HP and wrong answers | Player has 100 HP. The first wrong answer in a row costs nothing. Each further consecutive wrong answer costs 10 player HP. So 11 consecutive wrong answers (1 free + 10 × 10 HP) loses the fight. Any correct answer resets the consecutive-wrong count to zero. |
| Running out of questions | If all 20 questions are answered and the boss still has HP left, the attempt counts as a fail. |
| Retries | After losing, the player may retry at most 2 times — 3 attempts total per boss exam. The attempt counter is shown before and during the fight; a retry starts a fresh attempt with full HP on both sides, a fresh 1-hour timer and a newly randomised order. |
| XP — per answer | Each correct answer gives +5 XP. |
| XP — streak bonus | A streak is consecutive correct answers. From the third correct answer in a row onward, each correct answer in the streak also gives +5 bonus XP (the 3rd, 4th, 5th and so on each earn it). A wrong answer resets the streak to zero. |
| XP — win bonus | Defeating the boss gives a +100 XP win bonus, on top of per-answer and streak XP. Show all three lines separately on the victory panel. |
| Worked check | A perfect run: 10 correct answers end the fight at question 10 — 10 × 5 XP per answer = 50, streak bonus on answers 3–10 = 8 × 5 = 40, win bonus 100; total 190 XP before any quest-completion XP. Use this to test the implementation. |

## 5.5 Profile

The profile collects the student's game identity in three blocks:

- **Identity header:** hero sprite, name, level and rank line, total XP with the XP bar to the next level, and the current streak with the flame icon — all from GET /api/progress/me/.

- **Badges:** a grid of badge medallions (Section 6) with name and state under each. Earned badges are full colour; locked badges are greyscale at 45% opacity with the word Locked. Show every badge the API returns; do not invent badge types in the frontend.
- **Streak calendar:** specified here; the mock shows streak counts but no calendar, so build this component to this specification in the same tokens. A month grid, one cell per day, cells fully rounded or 13px radius: study days filled in the streak orange #ff9600 with white text, today outlined with a 2px primary-green border, missed and future days in the tint surface with muted text. Above the grid, the current streak and personal best as two small stat cards. Month navigation uses ghost buttons. The calendar renders the streak record returned by the API; it does not compute streaks itself.

### 5.6 Teacher view

The teacher view keeps the same chunky styling, so it reads as the same app, but its layout is a working roster, not a path. From top to bottom:

- **Class header:** class name in Fredoka, student count, and one line stating the teacher's powers in this app: create quests and quizzes, follow progress and scores.
- **Create row:** a primary Create Quest button and a ghost Create Boss Quiz button. The quest form collects title, subject, type (Assignment or Revision), difficulty and deadline. The quiz form collects subject, topic and question count, then generates the quiz through POST /api/ai/generate-quiz/ and shows the stored result for review before it goes live. Generation happens once per session: the generated quiz is cached server-side, so reopening, editing the class or assigning the same subject again reuses the cached quiz instead of calling the AI again, and the same cached quiz may be reused for a different class on the same subject. The teacher UI states when a quiz came from the cache (for example: Generated once · reused for XII IPA 2) and offers Regenerate only as a deliberate action, never as the default on every open. Neither form has any file field.
- **Class stats:** the shared stat card at teacher scale: active quests (with the Assignment / Revision split), average completion, boss quizzes (with how many are live), and average score on the last boss fight.
- **Student progress table:** a bordered, rounded table with a tint header row and columns Student (avatar initials disc and name), Quests (completed / total), XP, Last quiz (score) and Status. Status is a pill: On track (Revision-tag colours), Almost there (#fdf0dd background, #92580a text, #f3ddba border), Needs a nudge (#fdeaea background, #b02a2e text, #f6caca border). Data comes from GET /api/progress/. Rows are read-only — the teacher views progress and scores here and does not grade anything.
- **Quests you have set:** the quest cards for quests the teacher created, with a ghost View button, and edit and delete actions that call the quest endpoints (PATCH and DELETE /api/quests/{id}/).

## 6. Sprites and icons

The game feel comes from meaningful artwork, not from colour alone. Every icon below is an inline SVG in the app, drawn on a 24×24 grid with a 2.2 stroke and round caps and joins, exactly as in icons.svg. Do not replace subject icons with emoji, letters in circles or generic dots.

### 6.1 Subject icons

| Subject | Icon and placement |
|---|---|
| Matematika | Sigma (Σ). On a 46×46px tile in the subject blue, 13px radius, white icon at 26px, with an inset bottom shade (inset 0 -3px 0 rgba(0,0,0,.18)). On the dashboard path the same icon sits in the 52px circular path dot. |
| Biologi | Leaf, same tile and dot treatment in the subject green. |
| Fisika | Atom (nucleus with two orbit ellipses), same treatment in the subject amber. |
| Bahasa Inggris | Open book, same treatment in the subject pink. |

If a subject outside this list ever arrives from the API, fall back to the star icon on a neutral tile and flag it — do not silently reuse another subject's icon or colour.

### 6.2 Difficulty gems, streak flame and XP star

- **Difficulty is gems, not text alone:** three gem slots on every quest card. Easy fills 1 gem, Medium fills 2, Hard fills 3. Filled gems are #38bdf8; empty gems are #94a3b8 at 22% opacity. The difficulty word (Easy, Medium, Hard) always sits next to the gems, so difficulty never depends on colour alone. Difficulty also matches the mock's XP values: Easy 20 XP, Medium 50 XP, Hard 100 XP.
- **Streak is a flame:** the flame icon in #ff9600 wherever a streak appears — stat strip, profile, streak calendar header.
- **XP and level are a star and a bolt:** the star icon marks Level and badges; the lightning bolt marks XP amounts. Keep this pairing consistent on every screen.
- **Utility icons:** clock for due dates and the timer, check for completed states, crossed swords for boss fights, shield for the logo crest, target for the daily goal, plus for create actions.

### 6.3 Badge artwork

A badge is a 46px circular medallion in its own colour with a white 22px icon, the badge name in bold underneath and its state below that. The mock's set, for proportions: First Quest (amber #f59e0b, check icon), Boss Slayer (red #e5484d, sword icon), 7-Day Flame (orange #ff9600, flame icon) and Mastermind (blue #2563eb, star icon). Earned badges render at full colour. Locked badges render greyscale at 45% opacity with the label Locked. Badge definitions and earned state come from the API (the badges data behind GET /api/progress/me/); the frontend supplies only the medallion presentation, keyed by the badge's icon and colour fields.

### 6.4 Sprites

- **Granite Golem (boss):** the SVG sprite in boss-golem.svg — a slate stone body (#64748b with #7d8da3, #55677e, #475569 and #334155 facets), a purple crystal (#a855f7) on its head, glowing amber eyes (#fbbf24), moss-green strokes (#22c55e) and a gold crack (#fbbf24). Sizes: 132px standard, 168px in the boss fight arena, 84px small. One boss sprite is enough for this build; reuse the golem for every boss quiz, with the topic changing around it.
- **Student hero:** the small companion sprite from the mock (64px on the dashboard, with the same float animation). Keep it in the dashboard hero row and the profile header only.
- **Rendering rule:** sprites are inline SVG components, never raster images, so they stay sharp at every size and can be animated per part (the golem's eyes are a separate group for the blink, Section 7).

## 7. Motion

Motion is part of the specification, especially in the boss fight. Animate transforms and opacity only, keep every animation at or under the durations below, and honour the reduced-motion rule at the end of this section. Keyframe values are copied from the mock.

| Animation | Specification |
|---|---|
| Boss idle (bossFloat) | 2.6s ease-in-out, infinite. The golem bobs: translateY 0 and rotate -1deg at the ends, translateY -9px and rotate 1deg at the midpoint. A soft drop shadow (0 10px 8px rgba(0,0,0,.25)) grounds it. |
| Boss blink | 4s, infinite. The eye group scales vertically to 0.1 at 95% of the cycle and back by 100% — a blink roughly every 4 seconds. |
| Boss hit | 0.6s ease, played once per correct answer. Brightness flash (brightness 2, desaturated) settling through brightness 1.8 with a red hue shift, combined with a shake: translateX 0, -10px with rotate -4deg, 9px, - |

| | |
|---|---|
| | 6px, 0. Trigger it by re-adding the hit class so consecutive hits replay it. |
| Damage pop | 1s, forwards. A white pill with red weight-900 text (-10 DAMAGE, the fixed damage of one correct answer) sits centred at the top of the arena: it starts at opacity 0, scale 0.8, 8px low; it is fully visible by 20% and ends 24px higher, scale 1.1, faded out. |
| Boss defeat (crumble) | 1s, forwards, when HP reaches zero: the golem sinks translateY 26px, topples to rotate 72deg, fades to opacity 0.25 and turns greyscale. Then the victory panel replaces the fight controls. |
| Hero float | The student hero sprite uses the same float keyframes at 3s ease-in-out, infinite. |
| Button press | Primary buttons translate down 2px on press while their hard shadow shrinks from 4px to 2px. No other hover or press effects are needed; do not add ripples or glows. |
| Progress changes | XP and HP bar fills transition their width over about 0.4s so gains and damage read as movement, not jumps. |
| Quest reward pop | The quest detail's completion feedback (Section 5.3) reuses the damage pop's 1s rise-and-fade motion with the meaning reversed: a white pill with green weight-900 text (+100 XP · Quest complete) rises over the header card while the XP bar animates to its new width. No shake, no flash and no red anywhere — reward motion on study screens stays gentle, and the dramatic hit motion stays exclusive to the boss fight. |

Reduced motion: under the prefers-reduced-motion setting, disable the sprite float, the hero float and the blink entirely, as the mock does, and shorten the hit and defeat animations to a simple opacity change. A student who asks their system for less motion must still get every state change and feedback message — motion is decoration, never the only signal.

## 8. Responsive behaviour

One responsive React app serves phone and desktop browsers. The mock's own breakpoints are the floor of this specification; the layout rules below extend them to full screens.

| Width | Behaviour |
|---|---|
| Above 880px (desktop) | Content column centred at its maximum width (about 880px; teacher table about 1,100px). Stat strip in 4 columns. Answers in 2 columns. Top bar navigation. Path offsets at the full ±14px. |
| 560px and below | Stat strip drops to 2 columns. The teacher table hides the XP and Last quiz columns, keeping Student, Quests and Status. Badge grid wraps to 2 per row. |
| 480px and below | Answer buttons stack in 1 column. Path node offsets reduce to ±8px so nodes never clip the screen edge. Navigation moves to the bottom bar. The golem stays 168px in the arena — shrink the arena padding, not the boss. |

- Nothing may scroll horizontally at any width, and no text may be cut off inside cards or buttons.
- The boss fight must be fully playable one-handed on a phone: question, answers and Check Answer reachable without zooming.
- On the quest board the three filter groups wrap onto their own rows on phones without hiding any filter, and quest cards keep the full field set (icon, gems, XP, deadline, status) at phone width — drop the card's secondary menu into the detail rather than dropping a field. On the quest detail, the three reward tiles stack to one row of three compact tiles or a single column, the checklist rows stay at least 44px high for tapping, and the Complete Quest button stays full width.
- Test at a phone width (about 390px) and a desktop width (about 1,280px) before calling any screen done.

# 9. Frontend build rules

## 9.1 Components

Build at least these components, named for what they are; screens are compositions of them, not separate one-off markup:

| Component | Used on |
|---|---|
| AppShell / TopBar | Every screen: logo, navigation, avatar; bottom navigation on phones |
| StatStrip / StatCard / XpBar | Dashboard, profile header, teacher stats |
| QuestCard / PathNode | Quest board, dashboard path, teacher's set quests |
| SubjectTile / GemRow / Tag | Inside every quest card |
| QuestBoard / FilterChips | Quest board: type, subject and status filter groups, result count, card stack (Section 5.2) |
| QuestDetail | Quest detail screen: header, reward/deadline/difficulty tiles, description (Section 5.3) |
| QuestStepChecklist | Inside QuestDetail: tickable steps with progress bar; owns the in-progress state |
| RewardPop | Quest detail completion feedback; the gentle counterpart to the boss damage pop (Sections 5.3 and 7) |
| BossUnlockBanner | Quest detail, locked and unlocked states; its button routes to the linked BossFight (Section 5.3) |
| BossTeaser | Dashboard |
| BossSprite | Boss teaser and boss fight arena; owns the idle, blink, hit and defeat states |
| QuestQuiz | Inside QuestDetail / its own route: the regular 5-question quiz — question card, answers, instant feedback, result screen with pass at 80% and Retry (Section 5.3.1) |
| BossFight | The boss exam screen: arena, both HP bars, 1-hour timer, streak indicator, attempt / retry counter, 20 questions, win and defeat panels (Sections 5.4–5.4.1) |
| StreakIndicator / AttemptCounter | Boss exam header: current correct-answer streak and Attempt X of 3 with retries left |
| BadgeGrid / BadgeMedallion | Dashboard badge row, profile |
| StreakCalendar | Profile (specification in Section 5.5) |
| ProgressTable | Teacher view |
| QuestForm / QuizForm | Teacher create and edit flows; the student's Revision Quest form reuses QuestForm |

## 9.2 API mapping

The frontend talks to the backend only through the planned REST API from the Technical Specifications. The endpoints the UI uses are:

| UI need | Endpoint |
|---|---|
| Sign in, register, current user | POST /api/auth/login/, POST /api/auth/register/, GET /api/auth/me/ |
| Quest board lists and filters | GET /api/quests/ with type=assignment\|revision, subject and status parameters — one request per filter combination (Section 5.2) |
| Quest detail (description, steps, linked boss) | GET /api/quests/{id}/ — the response drives the header, description, steps checklist and the locked boss banner (Section 5.3) |
| Save step progress and quest status | PATCH /api/quests/{id}/ — ticking steps saves progress and the in-progress status; it never awards XP on its own |

| | |
|---|---|
| Complete a quest | PATCH /api/quests/{id}/ to the done status, then GET /api/progress/me/ for the confirmed XP total, level and streak shown in the reward pop |
| Boss unlock state and entry | GET /api/quizzes/ (and the quest detail's linked quiz) decides locked vs unlocked; the unlocked banner's button opens that quiz's boss fight |
| Create, edit, delete quests | POST /api/quests/, PATCH /api/quests/{id}/, DELETE /api/quests/{id}/ |
| Quiz lists and detail | GET /api/quizzes/, GET /api/quizzes/{id}/ |
| Create a quiz (teacher) | POST /api/quizzes/ |
| Regular quest quiz (load, submit, best attempt) | GET /api/quizzes/{id}/ for the quest's linked 5-question quiz; POST /api/quizzes/{id}/attempts/ submits an attempt and returns the score, the pass state at 80% and the best attempt — a pass is what ticks the quiz step (Section 5.3.1) |
| Start a boss exam attempt | POST /api/quizzes/{id}/attempts/ starts the attempt; the response carries the 20 questions in this attempt's randomised order with each question's options already shuffled, plus boss HP 100, player HP 100 and the 1-hour deadline |
| Submit and review a boss attempt | POST /api/quizzes/{id}/attempts/, GET /api/quizzes/{id}/attempts/{attempt_id}/ — the returned outcome (win / fail and why), per-answer XP, streak bonus XP and the +100 XP win bonus are the official values shown on the end panels; the attempts endpoint also enforces the 3-attempt total limit |
| Student XP, level, streak, badges | GET /api/progress/me/ |
| Teacher progress and scores | GET /api/progress/ |
| Generate a quiz or revision plan (teacher-triggered, server-side AI) | POST /api/ai/generate-quiz/, POST /api/ai/revision-plan/ — generation is once per session and cached server-side; the same cached quiz may serve a different class on the same subject, so a cache hit must not trigger a new generation call |

- **Server values win:** XP totals, level, streaks, badge states, quiz scores, boss and player HP, streak bonuses, attempt counts and boss outcomes are whatever the API returns, applying the exact numbers in Section 5.4.1 (boss 100 HP, 10 damage per correct answer, player 100 HP, first consecutive wrong free then 10 HP each, +5 XP per correct answer, +5 streak bonus XP from the third in a row, +100 XP win bonus). The frontend displays and animates them; it does not compute official values locally. Level follows the locked rule — a level-up every 200 XP — as returned by the progress endpoint.
- **Every list and form has three states:** loading, empty and error, all styled with the tokens in Section 3. Errors show a short friendly message and a retry action, never a raw status code or stack trace.
- **Forms validate before sending:** required fields, a deadline that is not in the past, and a difficulty chosen from Easy, Medium or Hard. Show validation messages under the field in the danger colour.
- **Accessibility basics:** icon-only buttons carry text labels for screen readers, the selected answer and the active tab are marked in state as well as colour, focus is visible on every interactive element, and difficulty is always shown as gems plus the difficulty word.

## 10. Acceptance checklist

- Tokens from Section 3 are defined once and used everywhere; no component hardcodes a colour, radius or shadow.
- Headings render in Fredoka and body text in Nunito, with working sans-serif fallbacks.
- The dashboard shows the winding path: dashed trail, alternating ±14px node offsets, 52px subject dots, stat strip, purple boss teaser and badge row.
- Quest cards show subject tile, gems plus difficulty word, XP pill, type tag and due date on every screen that lists quests.

- The quest board filters by type (Assignment / Revision), subject and status in combination through GET /api/quests/ parameters, shows a status pill (Not started / In progress / Done) on every card, and stays calm: no timer, HP bar, boss sprite or damage feedback on the board.
- Tapping any quest card opens the quest detail: description, deadline and XP tiles, and a tickable steps checklist that saves progress through the API and survives a reload.
- The quest detail has no timer and no HP bar; its single primary action is Complete Quest, available once any quiz step is passed and otherwise available regardless of the ordinary study steps.
- A quest's quiz step cannot be ticked by hand: it ticks only on a server-confirmed pass, and until then the quest stays incomplete and Complete Quest stays disabled with the reason shown.
- The regular quest quiz runs exactly 5 questions with no timer, gives instant feedback, passes at 80% (4 of 5), allows unlimited retries, and keeps the best attempt as the score of record.
- Completing a quest plays the reward pop, animates the XP bar to the API-confirmed total, flips the status to Done and locks the checklist — and shows no reward if the completion request fails.
- Completing a quest with a linked boss quiz replaces its locked banner with the Boss unlocked banner, whose Fight the Boss button opens that boss fight; a quest with no linked boss shows no banner.
- The boss exam plays end to end with the Section 5.4.1 numbers: 20 randomised questions with shuffled options per attempt, 1-hour countdown, boss 100 HP dropping 10 per correct answer, player 100 HP dropping 10 only from the second consecutive wrong answer onward, streak display, and idle bob, blink, hit flash-and-shake and damage pop on correct answers.
- All three boss losses work and are stated in words on the defeat panel: player HP reaches zero (11 consecutive wrong answers), the 1 hour expires, or the 20 questions run out with the boss alive; each loss offers a retry only while retries remain, up to 2 retries (3 attempts total).
- Boss XP is shown as separate lines and matches the rules: +5 XP per correct answer, +5 streak bonus XP per correct answer from the third in a row, and +100 XP win bonus on victory (a perfect 10-correct run totals 190 XP).
- Quiz generation is once per session and cached server-side: reopening or reassigning a quiz on the same subject does not regenerate it, and the cached quiz can be assigned to a different class on that subject.
- The profile shows level and XP, the badge grid with a visibly locked state, and the streak calendar built to Section 5.5.
- The teacher view has both create flows, the four class stats, the progress table with status pills, and the teacher's set quests — and no file upload, attachment or grading control anywhere in the app.
- No multiplayer, leaderboard or shop element exists in any screen, menu or placeholder.
- Every screen works at about 390px and about 1,280px with no horizontal scrolling, and all animations respect prefers-reduced-motion.